

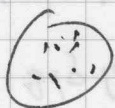
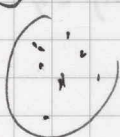
Clustering

Given $X = \{x_1, x_2, \dots, x_n\} \subseteq \mathbb{R}^d$, find a disjoint partition of $X = \bigcup_i X_i$, $\emptyset X_i \cap X_j = \emptyset, i \neq j$ s.t. point within X_i are 'similar' to each other and 'dissimilar' to point in $X_j, i \neq j$.

Remarks:

- The 'similarity' is usually measured by Euclidean distance $\|x_i - x_j\|_2$ or angle.
- If $\|x_i\| = 1$, then the similarity is measured by angles.
 $\|x_i - x_j\|_2^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle = 2(1 - \cos \theta) \Rightarrow \cos \theta = 1$, similar,
 $\cos \theta = -1$, not similar
- If not normalised, just use $\|x_i - x_j\|_2$.
- In practice, the number of partition is given (k).

Easy case



In this example of 'easy' clustering case, we have the following observation



Ob Obs: $\exists$ partition $X = \bigcup_i X_i$ s.t.

$$\max_{\substack{i,j \\ x_i, x_j \in \text{same } X_k}} \|x_i - x_j\|_2 \leq \min_{\substack{i,j \\ x_i, x_j \in \text{diff } X_k}} \|x_i - x_j\|_2$$

In this case, if we define distance matrix

$D_{ij} = \|x_i - x_j\|_2$, and $LUS < R < RUS$ of above, then $\exists$ permutation $\Pi: \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ and corresponding permutation matrix $P_\Pi = \begin{pmatrix} e_{\Pi(1)}^T \\ e_{\Pi(2)}^T \\ \vdots \\ e_{\Pi(n)}^T \end{pmatrix}$, $e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_{i \text{th}}$ s.t.

$D_\Pi = P_\Pi D P_\Pi^T$ which is "close" to a block diagonal matrix. i.e. within the diagonal block, $D_{ij} < R$, outside the block, $D_{ij} > R$.

Remarks: The above statement simply means that we can rearrange the point s.t. the point that 'close' are grouped together (clustering).

K-means Clustering

for $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$ and k partitions $X = X_1 \cup X_2 \dots \cup X_k = P$ and $\mu_1, \dots, \mu_k \in \mathbb{R}^d$, find the optimal $\{X_i\}$ and $\{\mu_i\}$ s.t.

$$\min_{\{X_i\}} \left(\min_{\{\mu_i\}} \sum_{i=1}^k \sum_{x_j \in X_i} \|x_j - \mu_i\|^2 \right)$$

Lemma: ~~$\mu_j^* = \frac{1}{|X_j|} \sum_{x_i \in X_j} x_i$~~ $\mu_j^* = \frac{1}{|X_j|} \sum_{x_i \in X_j} x_i$, where $\{\mu_j^*\} = \arg \min_{\{\mu_j\}} \sum_{j=1}^k \sum_{x_i \in X_j} \|x_i - \mu_j\|^2$ where $\{\mu_j^*\} = \arg \min_{\{\mu_j\}} \sum_{j=1}^k \sum_{x_i \in X_j} \|x_i - \mu_j\|^2$ for fixed $\{X_i\}$.

Proof: let $f = \sum_{j=1}^k \sum_{x_i \in X_j} \|x_i - \mu_j\|^2 = \sum_{j=1}^k \sum_{x_i \in X_j} (x_{i,p} - \mu_{j,p})^2$.

$x_{i,p}, \mu_{j,p}$ are the p th component of $x_i, \mu_j \in \mathbb{R}^d$.

$$\begin{aligned} \frac{\partial f}{\partial \mu_{j,p}} &= \sum_{j=1}^k \sum_{x_i \in X_j} \frac{\partial}{\partial \mu_{j,p}} (x_{i,p} - \mu_{j,p})^2 \delta_{ij} \delta_{p,q} \\ &= \sum_{j=1}^k \sum_{x_i \in X_j} 2(x_{i,p} - \mu_{j,p}^*) \delta_{ij} \\ &= \sum_{x_i \in X_k} 2(x_{i,p} - \mu_{k,p}^*) = 0 \end{aligned}$$

$$\Rightarrow \sum_{x_i \in X_k} x_{i,p} - |X_k| \mu_{k,p}^* = 0 \Rightarrow \mu_{k,p}^* = \frac{1}{|X_k|} \sum_{x_i \in X_k} x_{i,p}$$

$$\text{or } \mu_k^* = \frac{1}{|X_k|} \sum_{x_i \in X_k} x_i$$

Lemma: The optimal partition

$X_i^* = \{x_j \in X, \|x_j - \mu_i^*\|_2 \leq \|x_j - \mu_{i'}^*\|_2, \forall i \neq i'\}$, with tie broke arbitrarily, for fixed $\{\mu_j^*\}$.

proof: for a fixed $\{ \mu_i \}$, the objective.

$$\begin{aligned} \min_{\{X_i\}} \sum_{i=1}^k \sum_{x_j \in X_i} \|x_j - \mu_i\|_2^2 &= \min_{\{X_i\}} \sum_{i=1}^k \underbrace{\|x_1^i - \mu_i\|_2^2 + \|x_2^i - \mu_i\|_2^2 + \dots + \|x_{|X_i|}^i - \mu_i\|_2^2}_{x_j^i \in X_i} \\ &\quad + \underbrace{\|x_1^2 - \mu_2\|_2^2 + \dots + \|x_{|X_2|}^2 - \mu_2\|_2^2}_{x_j^2 \in X_2} \\ &\quad \vdots \\ &\quad + \underbrace{\|x_1^k - \mu_k\|_2^2 + \dots + \|x_{|X_k|}^k - \mu_k\|_2^2}_{x_j^k \in X_k}. \end{aligned}$$

Consider a point $x_j \in X_i$, and $\exists \mu_k$ s.t. $\|x_j - \mu_k\|_2^2 < \|x_j - \mu_i\|_2^2$.
 then a new partition with $X_i \leftarrow X_i - \{x_j\}$ and $X_k \leftarrow X_k + \{x_j\}$.
 will drop the objective value. Then, we can repeat this
 process until ~~for~~ $x_j \in X_i$ no such case can be found.
 Thus the objective is minimized when

$$X_i^* = \{x_j \in X, \text{ s.t. } \|x_j - \mu_i\|_2 \leq \|x_j - \mu_{i'}\|_2, \forall i' \neq i\}.$$

This two lemma suggest possible algorithm using
 alternating optimization.

Lloyd's algorithm.

Input $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$.

Initialize: Choose k random initial guesses for the.

$\mu_1(0), \dots, \mu_k(0)$ and correspondingly initial cluster assignment.

$\hat{C}_1(0), \dots, \hat{C}_k(0)$ by associate each x_j to the closest center.

For $t=1, 2, \dots$ till converge.

1. use current cluster $\hat{C}_1(t-1), \hat{C}_2(t-1), \dots, \hat{C}_k(t-1)$ update means.

$$\mu_j(t) = \frac{1}{|\hat{C}_j(t-1)|} \sum_{x_i \in \hat{C}_j(t-1)} x_i$$

2. Use the updated means to update the cluster assignments (assign x_i to its nearest cluster mean).

Convergence property of Lloyd's Algorithm.

Let $X = x_1 \cup \dots \cup x_k$

$$L(x_1, \dots, x_k) := \sum_{i=1}^k \sum_{x_j \in x_i} \|x_j - \frac{1}{|x_i|} \sum_{x_k \in x_i} x_k\|^2$$
 be the cost for the partition with optimal center $\hat{\mu}_j(t)$

Lemma 11:

Prob: In Lloyd's algorithm, we have.

$$L(x_1(t), \dots, x_k(t)) \geq L(x_1(t+1), \dots, x_k(t+1)).$$

Proof: ~~from the previous lemma, we know that.~~

~~suppose the opt~~ The optimal center updated is

$$\hat{\mu}_j(t+1) = \frac{1}{|x_j(t)|} \sum_{x_i \in x_j(t)} x_i.$$

and the updated partition assigning each x_i to the nearest $\hat{\mu}_j$, thus $\forall x_i \in x_j, \|x_i - \hat{\mu}_j\| \leq \|x_i - \hat{\mu}_{j'}\|, j' \neq j$.

~~this fact~~ ^{updated,}

This using lemma, we know $\{x_j\}$ is Lloyd's algorithm is

the minimizer of $L(x_1(t), \dots, x_k(t))$

$$\text{thus } L(x_1(t), \dots, x_k(t)) \geq L(x_1(t+1), \dots, x_k(t+1)).$$

Cor

Thm: Let $L(x_1(t), \dots, x_k(t))$ be the cost of Lloyd's alg.

thm. $\lim_{t \rightarrow \infty} L$ exist and attained in finite time.

more over, assume all partitions have distinct cost

or tie breaking rule chose properly, the partition it self converge in finite time, i.e $\exists T > 0$ s.t.

$$(x_1(T), \dots, x_k(T)) = (x_1(T'), \dots, x_k(T')) \quad \forall T \geq T.$$

Proof: the existence of limits ~~is~~ is shown as we have shown the loss drop monotonically and lower bound by 0. The finite termination as the search space for partition is always finite.

Complexity. at each iteration, the Lloyd's algorithm require

- $O(knd)$ time to assign n data x_i to k center by $\|u_i - x_i\|_2^2$.
- needs $O(nd)$ time to update mean center.

Remarks:

- The Lloyd's algorithm is guarantee to converge, however, ~~may~~ only to local minimizer.
- Random projection can be used to reduce time.
- Lloyd's ~~suffers from~~ sensitive to initialization and might converge to various local minimum.

Initialization Methods, k-means++

~~case~~ Lloyd's algo is initialization sensitive, one ~~prob~~ improvement is to initialize the ~~the~~ mean center as far as possible.

Pick $u_1 \in \{x_1, \dots, x_n\}$ uniformly random $T = \{u_1\}$

while $|T| \leq k$

choose $u_i \in X \setminus T$ s.t. maximize $\min_{t \in T} \|x - t\|_2$.

$T.append(u_i)$

However, this initialization might be sensitive to outliers (suppose an outlier is very far away from the rest).

The improvement to this is to add randomness, and choose the further point with higher probability.

K-means++ algo:

Pick $\mu_1 \in X$ uniformly random, $T = \{\mu_1\}$

while $|T| < k$

Pick ~~the~~ $\mu_{|T|+1} \in X \setminus T$ with prob ~~prop~~ $\propto \min_{t \in T} \|x - t\|_2$.

$T \leftarrow T \cup \{\mu_{|T|+1}\}$

Remarks: ~~K-means++ init~~. The init steps take $O(kn)$

as we need to find $\|x - t\|$ for $\underbrace{x_i \in X}_n$ and $\underbrace{t \in T}_k$, same as one iteration of Lloyd's.

- Obviously, this init. will not guarantee anything, however, it can be shown the init cost is "close" to the optimal cost by factor of $\log(k)$, on average.

prob: let $T \subseteq X$ be the init means in K-means++, and $T^* \subseteq X$ be the optimal k centers. The subset $x_1 \cup x_2 \cup \dots \cup x_k = X$ is chosen accordingly, then our cost function

$$L(T) := \min_{x_1 \cup x_2 \cup \dots \cup x_k = X} \sum_{\mu_i \in T} \sum_{x_j \in X_i} \|x_j - \mu_i\|_2^2.$$

then $E(L(T)) \leq C \cdot O(\log(k)) L(T^*)$, (C is some const).

Geometric Interpretation of ~~k~~-means Lloyd's

In the Lloyd's algorithm the partition for a fixed means is updated via.

$$X_i = \{x \in X : \|x - \mu_i\|_2 \leq \|x - \mu_j\|_2, j \neq i\}$$

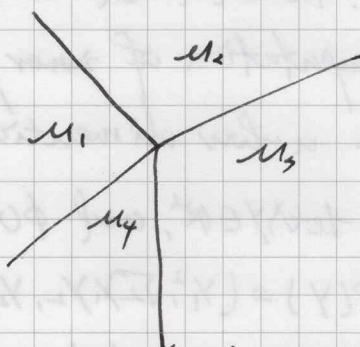
noticed that $\|x - \mu_i\|_2^2 \leq \|x - \mu_j\|_2^2, i \neq j$

$$\Rightarrow \|x\|_2^2 - 2\langle x, \mu_i \rangle \leq \|x\|_2^2 - 2\langle x, \mu_j \rangle$$

$$\Rightarrow -2\langle x, \mu_i - \mu_j \rangle \leq \|\mu_i - \mu_j\|_2^2 \text{ defines an}$$

half space separated by an hyper plane.

The Lloyd's algorithm clusters into intersection of $k-1$ half-planes



Remark: • It can be shown that the partitions in Lloyd's algorithm form Voronoi diagrams, and each is convex

• This means the Lloyd's alg essentially working with linear separable, convex clusters

Kernel trick.

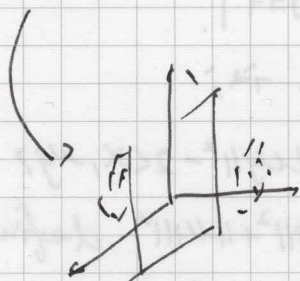
one way to deal with non-convex, non-linear cluster is to use dimension increasing, to make the original data to be separable more linear and convex in higher dimension

Example: Consider the following datas. in $\mathbb{R}^2$.



can create the map $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and

$$\phi(x_1, x_2) = \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}$$



However, boosting dimension will increase computation complexity. kernel-trick is a way to bypass this, by allowing implicitly computation of inner product of high-dimensional vector with a low dimensional function.

Example: let $x, y \in \mathbb{R}^2$, and $\phi(x) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$

$$\phi(y) = (y_1^2, \sqrt{2}y_1y_2, y_2^2)$$

$$\begin{aligned} \Rightarrow \langle \phi(x), \phi(y) \rangle &= x_1^2y_1^2 + 2\overset{x_1y_2y_1y_2}{\cancel{x_1x_2y_1y_2}} + y_2^2x_2^2 \\ &= (x_1y_1 + y_2x_2)^2 = \langle x, y \rangle^2 = k(x, y) \end{aligned}$$

Def kernel function. ^{symmetric}

$K: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, $\textcircled{A}$ suppose $\exists \phi: \mathbb{R}^d \rightarrow \mathbb{R}^n$ s.t.

$K(x, y) = \langle \phi(x), \phi(y) \rangle$, then K is called a kernel induced by ϕ .

Example: $\bullet K(x, y) = \langle x, y \rangle^r$

$$\bullet K(x, y) = (\langle x, y \rangle + c)^r$$

$$\bullet K(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right) \rightarrow \text{Gaussian kernel.}$$

Remarks: For the Gaussian kernel, the inducing map $\phi: \mathbb{R}^d \rightarrow \mathcal{H}$, $\mathcal{H}$ is Hilbert space, ~~with~~

Mercer's theorem:

Let $X \subseteq \mathbb{R}^d$ be an compact set, and, $K: X \times X \rightarrow \mathbb{R}$ be a continuous kernel, we ~~associate~~ define an linear integral operator associated with K on L^2 $T_K: L^2(X) \rightarrow L^2(X)$ as

$$T_K(f) := \int_X K(x, x') f(x') dx'$$

and suppose T_K is positive semidefinite.

$$\langle f, T_K(f) \rangle = \int_{X \times X} f(x) K(x, x') f(x') dx dx' \geq 0, \forall f \in L^2(X).$$

Then T_K has eigenfunction $\varphi_i, i=0, 1, \dots, \infty \in L^2$ and corresponding eigen value $\lambda_i \in \mathbb{R}^+$, s.t.

$$K(x, x') = \sum_{i=1}^{\infty} \lambda_i \varphi_i(x) \varphi_i(x')$$

Remark: This means we can view $K(x, x')$ as induced by a feature map $\varphi(x) := (\sqrt{\lambda_1} \varphi_1(x), \dots, \infty)$

k-means algorithm in kernel.

In Lloyd's algo, the partition is updated based on the distance to the ^{cluster} center.

$\|x_i - \frac{1}{|X_i|} \sum_{x_j \in X_i} x_j\|_2^2$, we first show this distance can be expressed as inner product.

$$\|X - \frac{1}{|X_i|} \sum_{x_j \in X_i} x_j\|_2^2 = \langle X, X \rangle - \frac{2}{|X_i|} \sum_{x_j, x_k \in X_i} \langle x_j, x_k \rangle + \frac{1}{|X_i|} \sum_{x_j \in X_i} \langle x_j, X \rangle.$$

~~the inner product can be re~~ if we use feature map

$\phi(x)$ to boost the dimension, then the distance in transformed dimension can be expressed in the kernel function

$$\|\phi(X) - \frac{1}{|X_i|} \sum_{x_j \in X_i} \phi(x_j)\|_2^2 = K(X, X) + \frac{2}{|X_i|^2} \sum_{x_j, x_k \in X_i} K(x_j, x_k) - \frac{2}{|X_i|} \sum_{x_j \in X_i} K(x_j, X).$$

Thus we modified Lloyd's algo as follows to include kernel.

Initialize: $X_1^{(0)} \cup X_2^{(0)} \dots \cup X_k^{(0)} = X$.

for $t = 1, \dots$ till converge:

update $X_i^{(t+1)} = \{x_j \in X : \|\phi(x_j) - \frac{1}{|X_i|} \sum_{x_k \in X_i} \phi(x_k)\|_2 \leq \|\phi(x_j) - \frac{1}{|X_i|} \sum_{x_k \in X_i} \phi(x_k)\|_2\}$

Remark: • The distance is calculated using the expression with kernel.

- The centers now are not explicitly calculated in feature space, but implicitly using kernels
- In practice, the kernels are pre-computed, each iteration
- Each iteration of Lloyd's takes $O(N^2)$